

A Scalable Hybrid Classifier for Music Genre Classification using Machine Learning Concepts and Spark

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Abstract—Music genre classification has been a challenging task in the field of Music Information Retrieval (MIR). Using a machine to automate this classification process is a more complex task. Audio feature extraction is the first important step of MIR. Essentia, a tool is used for the audio feature extraction. In this paper, a comparative study of the standard machine learning classifiers including K-Nearest Neighbor Classifier, Support Vector Machine, Naive Bayes Classifier, Neural Network, Quadratic Discriminant Analysis and Fuzzy Classifiers is undertaken in context of music genre classification. In this paper, a two phase hybrid classifier is proposed to overcome the problem of blurry classification of Pop, Rock and Electronic genres. It is observed that these genres are not classified accurately by above mentioned classifiers. The behavior of the proposed classifier on the standard music genre dataset GTZAN Genre Collection (1000 songs) and Free Music Archive (FMA) Dataset (approx. 20K songs) is investigated. The overall result of proposed approach on both the datasets is approximately 90% and 70% respectively.

Keywords—music information retrieval; essentia; machine learning concepts; feature extraction

I. INTRODUCTION

Digitalization of Music has made Internet a significant source of music pieces, with different on-line music consumption channels dedicated for spreading, distributing and commercializing music. Automatic procedures capable of handling large amounts of digital musical datasets are essential and Music Information Retrieval (MIR) has become an important research area. Genre is a subjective label assigned to music piece, which helps in categorizing music, to determine the style of the music. Humans have been the primary instruments in crediting genre-tags to songs. The way human assign music genre depend on his/her knowledge in that music-style, his/her perception of the music and ability to distinguish between the music-style genres whose boundary are blurry. This makes music genre classification a challenging job in the field of Music Information Retrieval (MIR). Automating of this classification process becomes a more complex task.

Audio feature extraction is an important task of audio processing, audio synthesis, audio effect design and music information retrieval [1]. Various audio feature extraction toolboxes are available like Essentia [2], Librosa [3],

Marsyas [4] etc. Among these tools, Essentia provides full function work-flow environment for low and high level audio features, providing audio input, preprocessing and analysis of output. Computation time taken by Essentia is very less compared to the other tools. In this paper, Essentia is used for the audio feature extraction [2].

Using artificial neural network classifier for Music Genre classification, Classical and Hip-hop are the best classified genres while Disco and Pop are not classified that well. This is because of the fact that the Disco and Pop are very broad categories and their boundaries are more blurry than Classical or Hip-hop.

To handle blurry genre boundaries, hybrid classifier constituting of two phases, is proposed.

In this paper, the proposed hybrid classifier on the standard music genre dataset GTZAN Genre Collection of 1000 songs and Free Music Archive (FMA) Dataset (~20K songs) is investigated. The proposed hybrid classifier was deployed on Spark platform to demonstrate scalability of the system.

The rest of the paper is organized as follows: Section 2 highlights the related work, Section 3 describes the methodology adopted. Section 4 discusses experimental setup and results, with Section 5 highlighting hybrid classification using Spark. The final section 6 summarizes the whole paper in terms of conclusion.

II. RELATED WORK

Music Genre Recognition (MGR) has been an active area of research since 1995 [5]. Various studies are presented based on the following aspects for MGR: methods for feature extraction and feature set used; classification algorithms applied for MGR; and some studies analyze system performance using certain figures of merit on particular benchmark datasets.

Music genre recognition utilizing temporal structure of a music piece has been explored by S. Rossignol et al. and S. Dubnov et al [6, 7]. S. Rossignol et al has proposed use of 9 features to distinguish between Speech, Singing Voice and Instrumental using k-Nearest Neighbors (k-NN) classifier, Gaussian Mixture Model (GMM) classifier and Neural Networks (NN) classifier. S. Dubnov et al demonstrated the use of texture and timbre properties for classifying into instrument families of brass, string and woodwind instruments. G. Tzanetakis et al [8], proposed use of 9

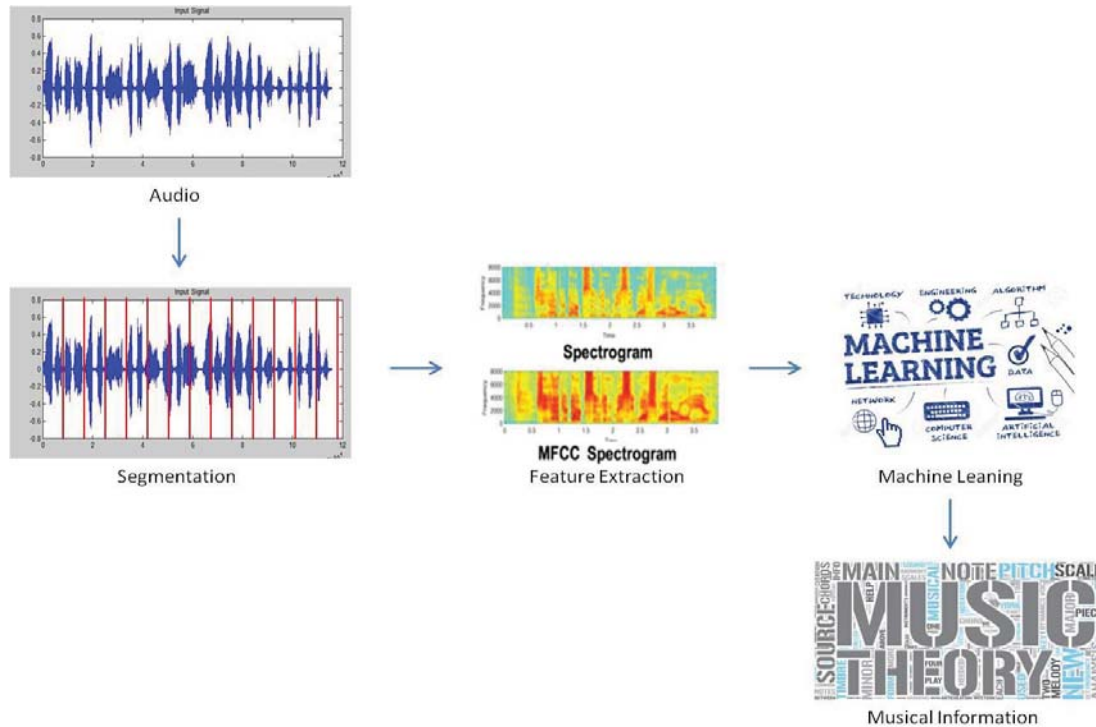


Figure 1. Music Information Retrieval System Architecture.

dimension feature vector, that include beat-related features, pitch-related feature and timbral texture features for automatic music genre recognition using k-Nearest Neighbors (k-NN) classifier and Gaussian Mixture Model (GMM) classifier. Changsheng Xu et al [9] proposed the use 5 features and Support Vector Machine (SVM) to classify music belonging to classical, rock, pop and jazz genres. Dharin Shah et al [10] have proposed use 16 features and Neural Network with softmax output.

According to [5], most of research work have considered private dataset, however there are many papers based on publicly available dataset like GTZAN [8], ISMIR2004 [11], RWC [12], etc... GTZAN dataset is the first benchmark dataset was made publicly available for MGR, and as a result continues to be the most used public MGR dataset [5]. On the GTZAN dataset the higher accuracy of approximately 84% is reported in [13].

III. METHODOLOGY

Fig. 1 depicts Music Information Retrieval (MIR) System Architecture. The input audio signal in segmented into frames of constant frame-size, hopping thought the audio signal with a constant hop-size, features are extracted on one frame at a time. The selected audio features are given as input to the Machine Learning models to get useful information about music.

A. Audio Feature Extraction

Audio feature extraction is one of the important tasks of MIR system architecture [1].

MPEG-7 “Multimedia Content Description Interface”, was initiated by ISO Motion Picture Experts in 1997, to develop an international MPEG-7 standard that would come

up with standardized descriptions of multimedia content and multimedia description systems [14]. Various audio feature extraction toolboxes are available like Essentia [2], Librosa [3], Marsyas [4] etc... Study and comparison of these toolboxes, in terms of number of audio features supported and their performances, is presented in [1]. Among the available audio extraction toolboxes, Essentia produce a strong largest range of audio features, and is the only tool box that provides 100% coverage of MPEG-7 audio descriptors. Computation time take by Essentia in processing 16.5 hours of audio is 4 minutes 12 seconds, is very less compared to tools like Marsyas that take 10 minutes and Librosa takes 1 hour 53 minutes [1]. In this paper, Essentia [2] is used for the audio feature extraction. Essentia tool provides large set of tonal, spectral, temporal and high-level music descriptors [15].

- Spectral Feature: features that describe the shape of power spectrum, computed from Short Time Fourier Transform of a signal frame. Example: centroid, spread, roll-off, skewness, complexity, crest, flux, energy.
- Temporal Feature: features that describe the tempo/rhythm, computed from each frame of audio signal. Example: Zero-crossing rate, loudness, onsets(attack-time).
- Tonal Feature: features that describe tonality i.e., arrangement of pitches and chords, for an audio signal. Example: chord change rate, chord strength, pitch salience.
- High Level Features: features like danceability, fade-ins, fade-outs.

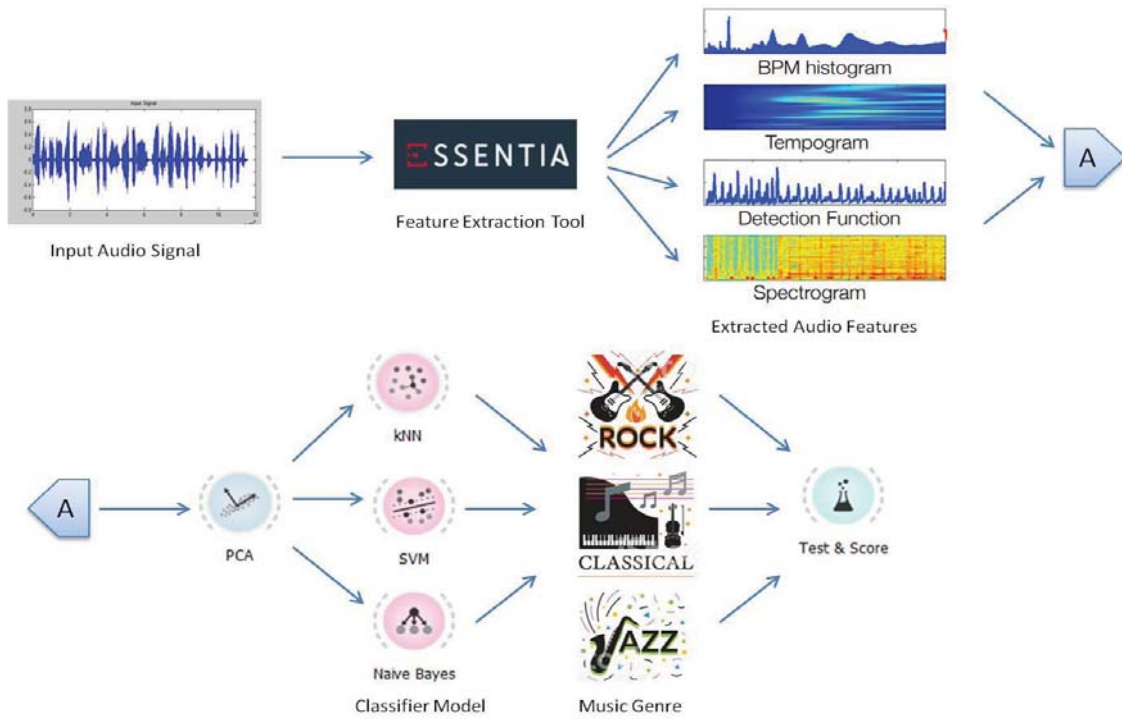


Figure 2. Music Genre Classification System Architecture.

B. Datasets

All sounds that are heard through our ears are pressure waves in the air. Recorded sounds can be digitized using an analog-to-digital converter (ADC). ADC takes an instantaneous amplitude sample of the analog waveform and converts it into digitalized sound. Samples are taken at regular intervals of time and the rate at which samples are measured is called the sampling rate (or sampling frequency). Bit-depth, is the number of bits used to describe each sample, also depends on the number of audio channels (mono, stereo). The product of the sample rate and number of bits per sample used to encode the audio is called the bit-rate. Audio stored in CD at 44,100 samples per second (or 44,100 Hz). CD audio is stereo i.e., 2 audio channels and 16 bits per channel. So, $44100 * 32 = 1411200$ (1411 kbit/s) is the bit-rate of uncompressed CD digital audio [16]. MP3 (MPEG I, layer 3) is a lossy compression technique that can reduce the digital audio file size, encodes this 1411 kbit/s data to 320 kbit/s or less [17]. In this paper, two different datasets are used:

1) *GTZAN*: This dataset contains 100 representative excerpts, for each of the 10 musical genres namely Blues, Classical, Country, Disco, Hip-hop, Jazz, Metal, Reggae, Pop and Rock. Each excerpt is 30 secs long taken from radio, compact disks, and mp3-encoded, stored as 22,050 Hz, 16-bit, mono audio files [8].

2) *FMA*: This dataset contains 106,574 excerpts, the audio for each excerpt is mp3-encoded, most of them with sampling rate of 44,100 Hz, 320 kbit/s (263 kbit/s on average) bit-rate and in stereo [18]. The audio is limited to 30 secs clips extracted from the middle of the tracks (or entire track if shorter than 30 secs). Unlike GTZAN dataset, this is not a

balanced dataset, with some genres like Experimental has 38,154 tracks, Electronic has 34,413 tracks, Rock has 32,923 tracks, Instrumental has 14,938 tracks and Blues has 1550 tracks.

Hence for this paper, approximately 2K tracks per genre including Blues, Classical, Electronic, Experimental, Folk, Hip-Hop, International, Jazz, Pop, Rock [18] are considered to have an approximately balanced dataset of ~20K tracks.

C. Music Genre Classification System Architecture

Fig. 2 shows the Music Genre Classification System Architecture proposed in this paper. The audio signal is given as an input to Essentia tool to extract audio features. From this tool, 68 audio descriptors were considered in total, describing global attributes of songs, and covering spectral, rhythmic, tonal and time-domain aspects of music track. Each track is represented as a data point with 68 features in feature space. Principle Component Analysis (PCA) was

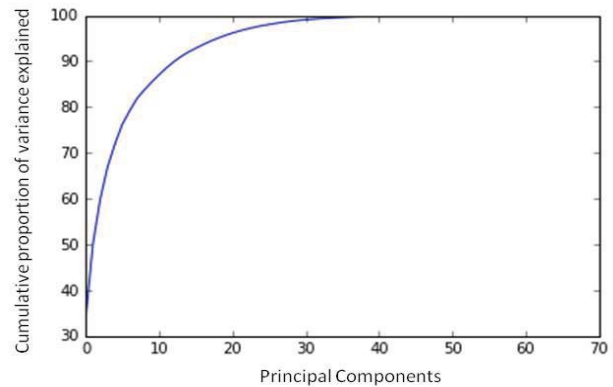


Figure 3. Principal Component Analysis Graph of GTZAN

applied to reduce the dimensions of the feature space.

Fig. 3 is the co-variance graph obtained with 68 features for 1000 tracks of GTZAN dataset.

The plot (Fig. 3) shows that 30 components results in variance close to approximately 98% are retained. In order words, using PCA, 68 features are reduced to 30 without compromising on explained variance. Therefore, for GTZAN, number of principle components are 30 [PC1 to PC30].

Similarly, Fig. 4 is the co-variance graph obtained with

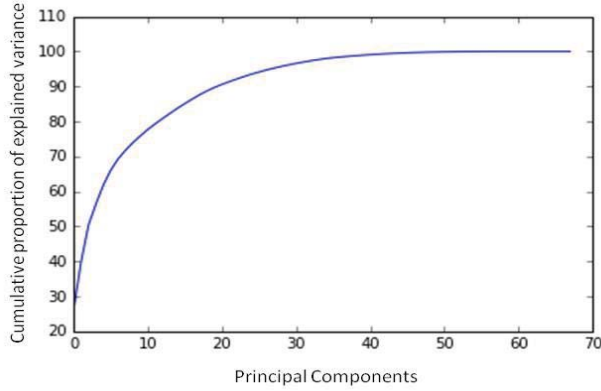


Figure 4. Principal Component Analysis Graph of FMA

68 features for approximately 20K tracks of FMA dataset. The plot (Fig. 4) shows that 30 components results in variance close to 97% are retained. Therefore, for FMA dataset, 68 features are reduced to 30 principle components [PC1 to PC30] to represent a track.

After selecting the 30 principle components, different Machine Learning models are trained and tested with 10-Fold Stratified samples, 9 Fold for training and 1 Fold for testing. The different Machine Learning Models used include K-Nearest Neighbor Classifier, Linear Support Vector Machine, Neural Network Classifier, Naive Bayes Classifier, Quadratic Discriminant Analysis.

The accuracy of various Machine Learning models on GTZAN dataset is as shown in Table I. It can be seen that Neural Network gives a very good classification accuracy of 82.4% and Naive Bayes has least accuracy.

TABLE I ACCURACY OF DIFFERENT ML CLASSIFIERS

	GTZAN
Nearest Neighbors	70.50%
Linear SVM	79.90%
Neural Net	82.40%
Naive Bayes	69.20%
Quadratic Discriminant Analysis	79.20%

The confusion matrix of Neural Net Classifier for GTZAN dataset is shown in Fig. 6. It can be seen that, Classical and Hip-hop are the best classified genres while Disco, Reggae and Pop are not classified that well. This is due to the reason that the genre boundaries Disco, Reggae and Pop are not clear cut as Classical or Hip-hop i.e., the boundaries between genre are blurry. Therefore, this paper proposes Hybrid Classification approach, a 2-phase approach. In stage-1 Neural Network Classifier is applied, and in stage-2 the miss-classified instances of stage-1 are split into set-1 of instances whose genre boundary are more clear and set-2 of instances whose genre boundaries are fuzzy. Further Neural Network Classifier is applied to the set-1 instances and FURIA: Fuzzy Unordered Rule Induction Algorithm [19] is applied to set-2. The block diagram of proposed hybrid classifier is shown in Fig. 5. The result of applying stage-2 of Hybrid approach for GTZAN dataset are shown in Fig. 7 and Fig. 8. The result of applying stage-2 of Hybrid approach for FMA dataset are shown in Fig. 10 and Fig. 11.

IV. EXPERIMENTAL SETUP AND RESULTS

A. GTZAN Dataset

The results of applying Neural Network Classifier for GTZAN dataset (1000 songs) are as shown in Table II and Fig. 6.

The accuracy of stage-1 (NN) on GTZAN dataset is 82.4%, i.e., 824 songs are classified correctly. Then, 176 songs that are miss-classifies in stage-1 are split into, 80 songs belonging to Disco, Reggae and Pop for FURIA Classifier [19] and 96 songs belonging to other 7 genres for stage-2 NN Classifier. From the confusion matrix it can be seen that 33 songs are correctly classified from FURIA (Fig. 8) and 26 songs are correctly classified from stage-2 NN (Fig. 7). Total correctly classified songs are $824 + 26 + 33 = 883$ songs, giving an accuracy of 88.3%.

TABLE II CLASSIFICATION METRIC OF NEURAL NETWORK FOR GTZAN

	precision	recall	f1score	support
Classical	0.93	0.94	0.94	100
Country	0.85	0.91	0.88	100
Hip-hop	0.85	0.89	0.87	100
Disco	0.72	0.69	0.70	100
Blues	0.88	0.82	0.85	100
Reggae	0.82	0.74	0.78	100
Jazz	0.76	0.84	0.80	100
Metal	0.84	0.82	0.83	100
Rock	0.80	0.82	0.81	100
Pop	0.79	0.77	0.78	100
avg / total	0.82	0.82	0.82	100

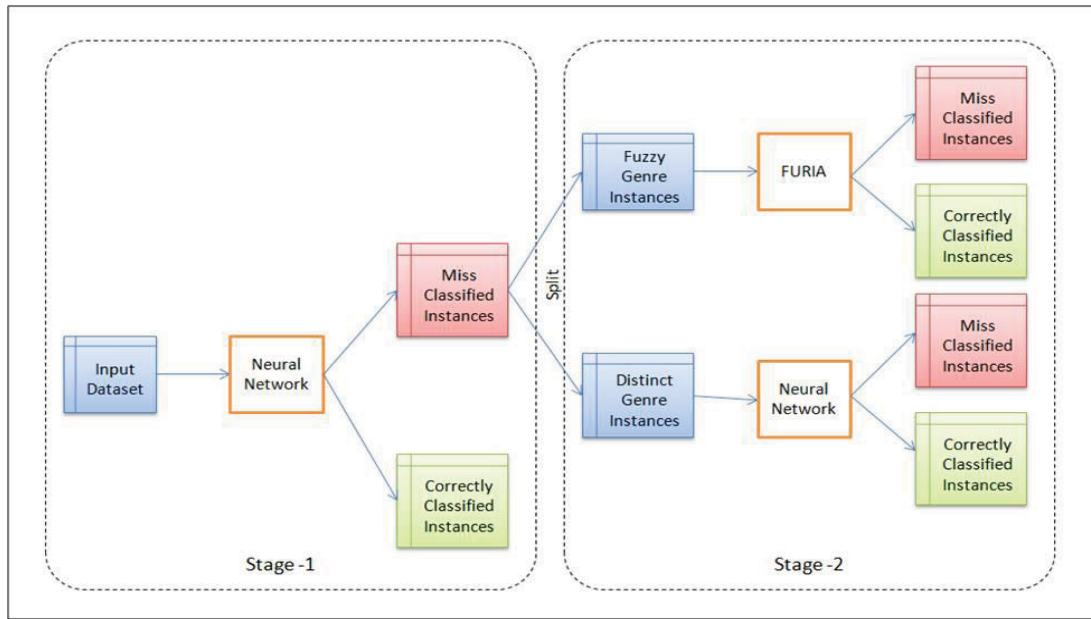


Figure 5. Block Diagram of Hybrid Classifier

classical	94	1	0	1	0	1	2	0	0	1
country	0	91	1	5	3	0	0	0	0	0
hip-hop	1	0	89	3	1	4	0	2	0	0
disco	1	6	5	69	4	5	1	3	2	4
blues	0	6	0	6	82	1	3	2	0	0
reggae	1	1	3	5	1	74	5	2	0	8
jazz	4	1	0	0	1	3	84	1	4	2
metal	0	1	5	1	1	0	2	82	8	0
rock	0	0	0	0	0	1	7	5	82	5
pop	0	0	2	6	0	1	6	1	7	77
	classical	country	hip-hop	disco	blues	reggae	jazz	metal	rock	pop

Figure 6. Confusion Matrix of Neural Network (GTZAN)

classical	0	0	0	2	3	0	1
country	0	1	3	5	0	3	0
hip-hop	1	1	4	2	0	3	0
blues	0	4	1	6	3	3	1
jazz	3	0	0	2	4	1	6
metal	0	0	3	2	1	6	6
rock	0	0	1	1	5	6	5
	classical	country	hip-hop	blues	jazz	metal	rock

Figure 7. Confusion Matrix of Neural Network Stage-2 (GTZAN)

disco	18	8	5
reggae	12	5	9
pop	8	5	10
	disco	reggae	pop

Figure 8. Confusion Matrix of FURIA Stage-2 (GTZAN)

B. FMA Dataset

The result of applying Neural Network Classifier for FMA dataset (19503 songs) are as shown in Table III and Fig. 9.

TABLE III. CLASSIFICATION METRIC OF NEURAL NETWORK FOR FMA

	precision	recall	f1score	support
Blues	0.69	0.60	0.64	1550
Classical	0.86	0.90	0.88	1967
Electronic	0.42	0.26	0.32	2000
Experimental	0.25	0.19	0.21	1996
Folk	0.25	0.17	0.20	1988
Hip-Hop	0.49	0.66	0.56	1991
International	0.43	0.51	0.46	1993
Jazz	0.40	0.54	0.46	2003
Pop	0.40	0.36	0.38	2003
Rock	0.53	0.62	0.57	2002
avg / total	0.47	0.48	0.47	19503

The accuracy of stage-1 (NN) on FMA dataset is 47.9%, i.e., 9332 songs are classified correctly. Then, 10,171 songs that are miss-classifies in stage-1 are split into, 4402 songs belonging to Electronic, Folk and Pop for FURIA Classifier and 5768 songs belonging to other 7 genres for stage-2 NN Classifier. From the confusion matrix it can be seen that 1778 songs are correctly classified from FURIA (Fig. 11) and 2534 songs are correctly classified from stage-2 NN (Fig. 10). Total correctly classified songs are $9332 + 2534 + 1778 = 13644$ songs, giving an accuracy of 69.95%.

V. HYBRID CLASSIFICATION ON SPARK

Training Machine Learning (ML) classifiers on a larger dataset takes longer time. One of the ways to reduce the training time, is to train ML classifiers with subset of data. Splitting dataset into 3-Stratified subsets and applying Neural Network classifier (non-hybrid approach), accuracy of FMA Dataset has reduced to 45.83% (NN) from 47.90%. Therefore, using smaller dataset doesn't seem to benefit

blues	932	160	3	2	1	24	62	54	62	250
classical	95	1768	1	2	1	2	0	0	4	94
electronic	8	1	529	279	223	278	173	273	144	92
experimental	6	2	232	376	481	205	230	236	130	98
folk	5	1	232	534	341	202	232	229	132	90
hip-hop	5	1	79	75	68	1314	134	130	115	70
international	9	5	63	83	88	220	1013	235	174	103
jazz	2	5	47	64	57	183	201	1089	233	122
pop	130	32	19	74	76	181	246	330	729	186
rock	157	83	56	17	11	69	83	166	118	1242
	blues	classical	electronic	experimental	folk	hip-hop	international	jazz	pop	rock

Figure 9. Confusion Matrix of Neural Network (FMA)

blues	388	20	28	9	64	28	81
classical	71	37	1	1	0	0	89
experimental	10	0	1115	63	181	187	64
hip-hop	22	0	302	75	130	104	44
international	32	0	362	41	320	156	69
jazz	16	0	317	29	136	363	53
rock	110	29	136	19	105	125	236
	blues	classical	experimental	hip-hop	international	jazz	rock

Figure 10. Confusion Matrix of Neural Network Stage-2 (FMA)

folk	690	862	105
pop	240	984	50
electronic	532	835	104
	folk	pop	electronic

Figure 11. Confusion Matrix of FURIA Stage-2 (FMA)

music genre classification. Another way is to split dataset into independent chunks and perform ML classifier's training in parallel manner. Hence, to reduce ML classifier training time without compromising on accuracy, music genre classification is performed on Spark platform [20].

For this paper, music genre classifier on Spark was deployed in multi-thread mode on single-node. Among various cluster managers (Spark's own standalone cluster manager, Mesos or YARN) supported by Spark [20], standalone cluster manager was used to form a cluster of 3-executors.

The 30 principal components were used for training 4-layer Multi Layer Perceptron (MLP) [21] containing 30 input layer neurons, 40 neurons in hidden layer-1, 15 neurons in hidden layer-2 and 10 neurons in output layer. The stage-2 FURIA was run on WEKA's [22] distributed Spark flow.

A. GTZAN Dataset

The result of applying Hybrid Classifier on Spark for GTZAN dataset (1000 songs) is as shown in Table IV. Runtime of 10 fold cross validation and Accuracy (total number of correctly classified songs) are shown.

TABLE IV. RESULTS OF GTZAN - HYBRID CLASSIFIER ON SPARK

	Run Time (s)			Accuracy		
	local	local-2	local-3	local	local-2	local-3
Stg-1 MLP	48.96	41.61	33.92	743	723	743
Stg-2 MLP	35.61	26.45	23.84	81	82	67
FURIA	3	2	2	76	69	66
Total	87.57	70.06	59.76	900	874	876

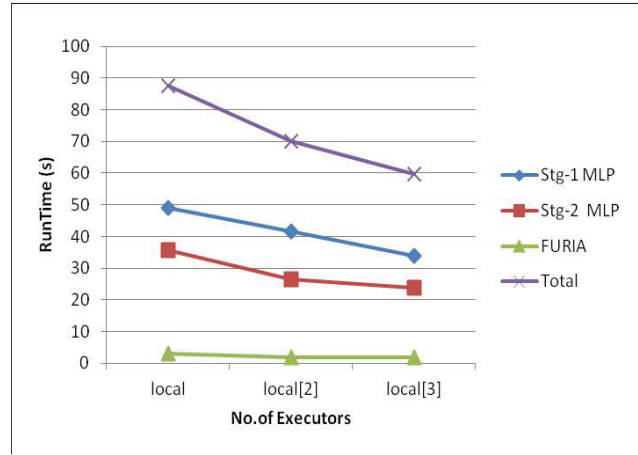


Figure 12. Runtime of GTZAN Dataset

For stage-1 Spark MLP, 10 genres are considered. Misclassified songs from stage-1 are split into 2 subsets, songs belonging to Disco, Pop and Reggae for FURIA and songs belonging to other 7 genres for stage-2 Spark MLP. Total accuracy of 90%, 87.4% and 87.6% are obtained for 1, 2 and 3 executors respectively. Fig. 12 shows run time of hybrid classifier on GTZAN dataset, total run time is decreasing as the number of executors increases, while accuracy is not affected much as shown in Fig. 14.

B. FMA Dataset

The result of applying Hybrid Classifier on Spark for FMA dataset (19506 songs) is as shown in Table V. Runtime of 10 fold cross validation is shown on the LHS and Accuracy (total number of correctly classified songs) are shown on the RHS. For stage-1 Spark MLP, 10 genres are considered. Misclassified songs from stage-1 are split into 2 subsets, songs belonging to Electronic, Folk and Pop for FURIA and songs belonging to other 7 genres for stage-2 Spark MLP. Total accuracy of 70.48%, 70.26% and 70.56% are obtained for 1, 2 and 3 executors respectively.

Fig. 13 shows run time of hybrid classifier on FMA dataset, total run time is decreasing as the number of executors increases, while accuracy is not affected much as shown in Fig. 14.

TABLE V. RESULTS OF FMA - HYBRID CLASSIFIER ON SPARK

	Run Time (s)			Accuracy		
	local	local-2	local-3	local	local-2	local-3
Stg-1 MLP	408.53	255.44	208.36	10189	10133	10025
Stg-2 MLP	171.04	113.24	91.39	2000	1864	2014
FURIA	67	39	29	1556	1706	1723
Total	646.57	407.68	328.75	13745	13703	13762

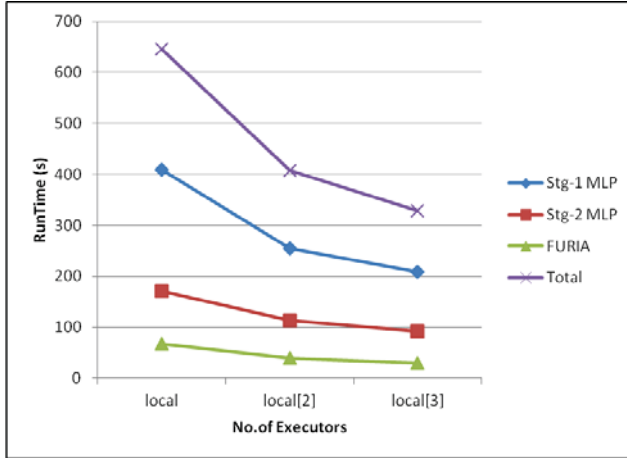


Figure 13. Runtime of FMA Dataset

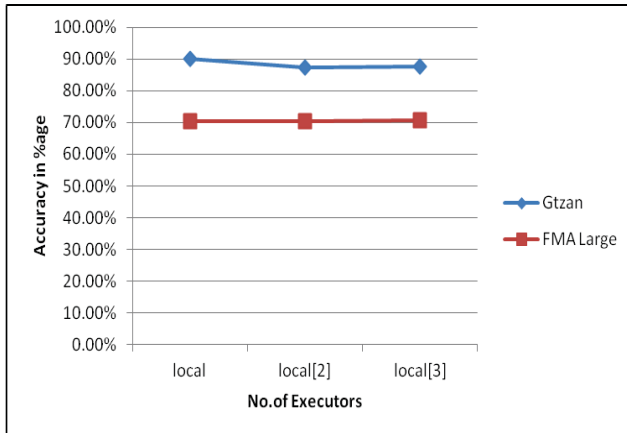


Figure 14. Accuracy of Hybrid Classifier on Spark

VI. CONCLUSIONS

The complexity of music genre classification increases as size of dataset increases and genre categories increases. In this paper, an investigation of using Essentia audio feature extraction tool and Neural Network for Music Genre Classification was done and evaluated on GTZAN and FMA Datasets. Blurriness in the Music Genre boundaries has driven proposal of Hybrid Classifier for Music Genre Classification. Results show that use of hybrid classifier

gives better classification accuracy. Further, using Spark framework, it could be seen that the runtime of Music Genre classification for larger dataset (FMA Dataset) is horizontally scalable. Music Genre classification accuracy of 90% obtained for GTZAN dataset with Hybrid classification method on Spark is better than earlier proposed methods, to the best of our knowledge.

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